TE4 Worked Examples: Solving Right Triangles

Recording companion for the TE4 CheckIt support video — Math 172, Santa Ana College

How to use this document

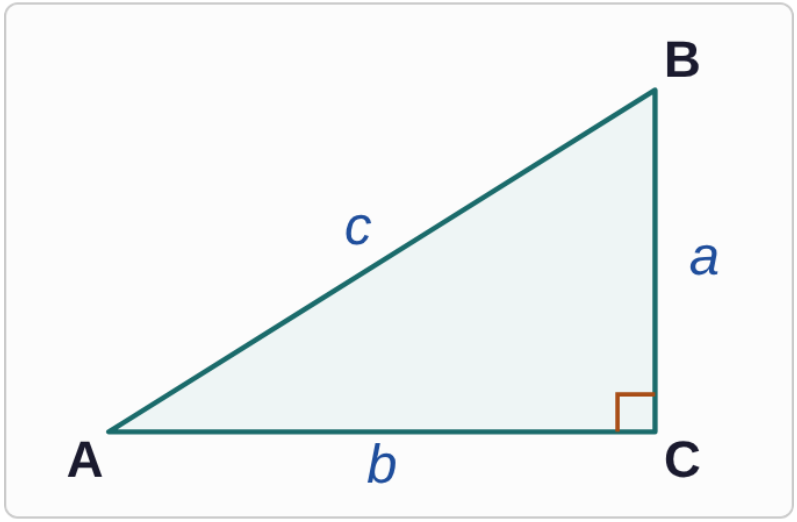
- Each worked example is on its own page, in the same order as the Try It tabs in the interactive explainer, with the same numbers.
- The right-hand third of every page is intentionally blank. It is space for handwriting or ink notation while you narrate (for example, after inserting this PDF as a printout in OneNote).
- Say it at the start of the video: your CheckIt problem will have different numbers than the examples here. The reasoning is what transfers.

The one question behind every example

Which sides and angles do I know, and which tool connects them to what I want? Label first: side a is across from angle A, side b across from B, and c, the hypotenuse, across from the right angle. With two sides, use the Pythagorean Theorem for the third side and inverse trig for an angle. With an angle and a side, choose sine, cosine, or tangent by which sides are involved. The two acute angles add to 90 degrees. Keep the calculator in degree mode, and round only at the end.

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Example 1: Label 1



Plotted data: A right triangle. Vertex A is at the bottom left, C at the bottom right, and B at the top right. A small square marks the corner at C, so the angle at C is 90°. Side b is the bottom side, joining A and C. Side a is the vertical side, joining C and B. Side c is the slanted side, joining A and B.

Question

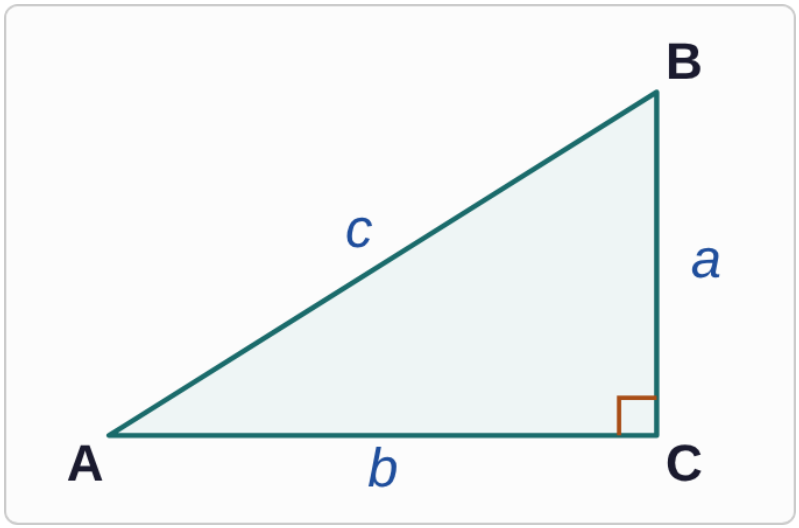
Which side is the hypotenuse?

- **A.** Side a
- **B.** Side b
- **C.** Side c
- **D.** It changes depending on angle A.

Answer and reasoning

The hypotenuse is the side across from the right angle. The right angle is at C, and the side that does not touch C is c, joining A and B. It is always the longest side. **The hypotenuse is c.**

Example 2: Label 2



Plotted data: A right triangle. Vertex A is at the bottom left, C at the bottom right, and B at the top right. A small square marks the corner at C, so the angle at C is 90°. Side b is the bottom side, joining A and C. Side a is the vertical side, joining C and B. Side c is the slanted side, joining A and B.

Question

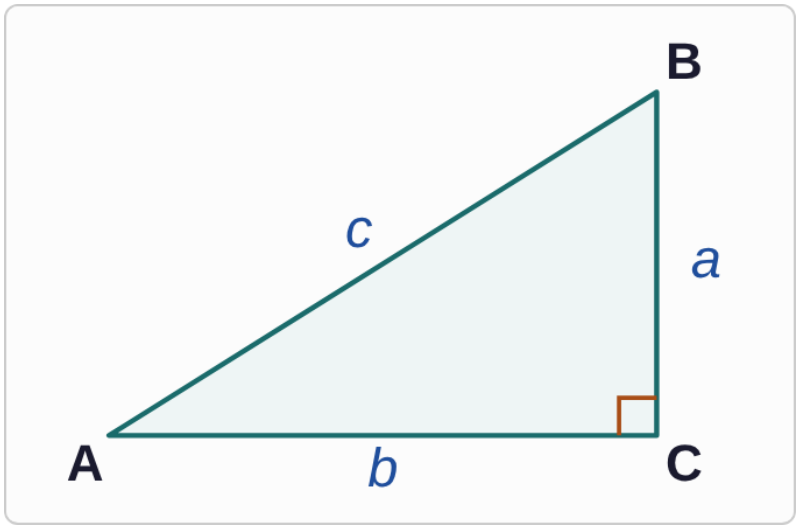
Which side is the opposite side for angle A?

- **A.** Side b
- **B.** Side a
- **C.** Side c
- **D.** Angle A has no opposite side.

Answer and reasoning

The opposite side for A is the side that does not touch A: side a, joining B and C. Side b touches A, so it is the adjacent side, and c is the hypotenuse. That is why the lowercase letter matches: side a is across from angle A. **The opposite side for angle A is a.**

Example 3: Label 3



Plotted data: A right triangle. Vertex A is at the bottom left, C at the bottom right, and B at the top right. A small square marks the corner at C, so the angle at C is 90°. Side b is the bottom side, joining A and C. Side a is the vertical side, joining C and B. Side c is the slanted side, joining A and B.

Question

Which ratio equals sin *A*?

- **A.** *b/c*
- **B.** *a/b*
- **C.** *c/a*
- **D.** *a/c*

Answer and reasoning

Sine is opposite over hypotenuse. For angle *A*, the opposite side is *a* and the hypotenuse is *c*. The ratio *b/c* is cos *A*, and *a/b* is tan *A*. **sin *A*=*a*/*c*.**

Example 4: Label 4

$$C=90^\circ, \quad A=50^\circ, \quad c=8$$

Question

Which equation finds side a?

- **A.** $a=8\sin 50^\circ$
- **B.** $a=8\cos 50^\circ$
- **C.** $a=8/(\sin 50^\circ)$
- **D.** $a=8\tan 50^\circ$

Answer and reasoning

Side a is opposite angle A, and $c=8$ is the hypotenuse. Opposite and hypotenuse means sine: $\sin 50^\circ = a/8$. Multiply both sides by 8. Cosine would find b, the adjacent side. ***a=8*sin *50°*.**

Example 5: Angle and Side 1

$$C=90^\circ,\quad A=35^\circ,\quad c=12$$

Question

Find side a.

A. about 9.8
B. about 6.9
C. about −5.1
D. about 20.9

Answer and reasoning

Side a is opposite A, and c is the hypotenuse, so use sine: $a=12\sin 35^\circ\approx 6.88$. The answer 9.8 uses cosine, which gives b. The answer -5.1 comes from a calculator in radian mode; a side length can never be negative. The answer 20.9 divides by sine instead of multiplying. Then $B=90^\circ-35^\circ=55^\circ$. $\boldsymbol{a\approx 6.9}$**.**

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Example 6: Angle and Side 2

$$C=90^\circ, \quad B=62^\circ, \quad a=7$$

Question

Find side b.

A. about 3.7
B. about 6.2
C. about 14.9
D. about 13.2

Answer and reasoning

This time the angle is B. Side b is opposite B, and side a touches B, so it is adjacent. Opposite and adjacent means tangent: $\tan 62^\circ = b/7$, so $b = 7\tan 62^\circ \approx 13.17$. The answer 3.7 uses angle $A=28^\circ$ instead, and 14.9 is the hypotenuse c. ***b*≈ 13.2.**

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Example 7: Angle and Side 3

$$C=90^\circ, \quad A=20^\circ, \quad b=15$$

Question

Find side c.

A. about 16.0
B. about 14.1
C. about 43.9
D. about 5.1

Answer and reasoning

Side b is adjacent to A, and c is the hypotenuse, so use cosine: $\cos 20^\circ = 15/c$. The unknown is on the bottom, so $c = 15/(\cos 20^\circ) \approx 15.96$. The answer 14.1 multiplies instead of dividing, and it is shorter than b, which cannot happen for a hypotenuse. The answer 43.9 uses sine. ***c≈ 16.0*.**

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Example 8: Angle and Side 4

$$C=90^\circ, \quad A=27^\circ, \quad a=11$$

Question

Find side c.

A. about 5.0
B. about 21.6
C. about 24.2
D. about 12.3

Answer and reasoning

Side a is opposite A, so use sine: $\sin 27^\circ = 11/c$, and $c = 11/(\sin 27^\circ) \approx 24.23$. The answer 21.6 is side b, from tangent. The answer 5.0 multiplies instead of dividing, and 12.3 uses cosine. The other angle is $B=63^\circ$. ***c≈ 24.2*.**

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Example 9: Two Sides 1

$$C=90^\circ, \quad a=8, \quad b=15$$

Question

Find side c.

A. 23
B. 17
C. about 12.7
D. 7

Answer and reasoning

Both legs are known, so add their squares: $c^2=8^2+15^2=64+225=289$, and $c=\sqrt{289}=17$. Adding the sides themselves gives 23. The answer 12.7 subtracts the squares, which is only right when c is one of the known sides. ***c=17*.**

Example 10: Two Sides 2

$$C=90^\circ, \quad a=20, \quad c=29$$

Question

Find side b.

A. about 35.2
B. 9
C. 21
D. 3

Answer and reasoning

Here $c=29$ is the hypotenuse, and b is a leg, so subtract: $b^2=29^2-20^2=841-400=441$, and $b=21$. The answer 35.2 adds the squares, which would make a leg longer than the hypotenuse. The answer 9 subtracts the sides without squaring. ***b=21*.**

Example 11: Two Sides 3

$$C=90^\circ, \quad a=8, \quad b=15$$

Question

Find angle A, to the nearest tenth of a degree.

A. about 61.9°
B. about 0.5°
C. about 32.2°
D. about 28.1°

Answer and reasoning

For angle A, side $a=8$ is opposite and $b=15$ is adjacent, so $\tan A=8/15$. Use inverse tangent: $A=\tan^{-1}(8/15)\approx 28.07^\circ$. The answer 61.9 is angle B, from flipping the ratio. The answer 0.5 is the ratio itself, not an angle. The answer 32.2 uses inverse sine with the wrong sides. $\boldsymbol{A\approx 28.1^\circ}$**.**

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Example 12: Two Sides 4

$$C=90^\circ, \quad b=24, \quad c=25$$

Question

Find angle A, to the nearest tenth of a degree.

A. about 16.3°
B. about 73.7°
C. about 43.8°
D. about 0.3°

Answer and reasoning

Side b is adjacent to A, and c is the hypotenuse, so $\cos A = 24/25$ and $A = \cos^{-1}(24/25) \approx 16.26^\circ$. The answer 73.7 uses inverse sine, which gives angle B. The answer 43.8 uses inverse tangent with the hypotenuse. The answer 0.3 comes from radian mode. $\boldsymbol{A \approx 16.3^\circ}$.

Example 13: Slip 1

$$a = 10\cos 40^\circ \approx 7.7$$

Question

In a right triangle, $C = 90^\circ$, $A = 40^\circ$, and $c = 10$. A student wrote this line. What is the mistake, if any?

A. There is no mistake.
B. Side a is opposite angle A, so it needs sine, about 6.4.
C. The 10 should be divided by the cosine of 40, about 13.1.
D. The angle should be 50 degrees, with sine: about 7.7.

Answer and reasoning

Side a is opposite A, and c is the hypotenuse, so the ratio is sine: $a = 10\sin 40^\circ \approx 6.4$. Cosine gives the adjacent side, $b \approx 7.7$. **The correct value is $a \approx 6.4$.**

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Example 14: Slip 2

$$b=\sqrt{(6^2+10^2)}\approx 11.7$$

Question

In a right triangle, $C=90^\circ$, $a=6$, and $c=10$. A student wrote this line. What is the mistake, if any?

A. The 6 and 10 should be multiplied first: about 7.7.
B. No square root was needed, so b is 136.
C. Side c is the hypotenuse, so subtract: b is 8.
D. There is no mistake.

Answer and reasoning

The hypotenuse is $c=10$, and it must be the longest side, so b cannot be 11.7. Subtract: $b^2=10^2-6^2=64$, and $b=8$. **The correct value is $b=8$.**

Example 15: Slip 3

$$c=13$$
$$A=\tan^{-1}(5/12)$$
$$\approx 22.6^\circ$$

Question

In a right triangle, $C=90^\circ$, $a=5$, and $b=12$. A student wrote these lines. What is the mistake, if any?

A. Angle A should use inverse sine of 5 over 12.
B. The hypotenuse should be 17, from 5 plus 12.
C. There is no mistake.
D. The ratio should be flipped to 12 over 5.

Answer and reasoning

$5^2+12^2=169=13^2$, so $c=13$. For angle A, side 5 is opposite and 12 is adjacent, so tangent is right: $A\approx 22.6^\circ$, and $B\approx 67.4^\circ$. **There is no mistake.**

Example 16: Slip 4

$$a = 10\sin 30^\circ \approx -9.9$$

Question

In a right triangle, $C = 90^\circ$, $A = 30^\circ$, and $c = 10$. A student wrote this line. What is the mistake, if any?

A. Side a needs cosine here: about 8.7.
B. There is no mistake.
C. The 10 should be divided by the sine: a is 20.
D. The calculator is in radian mode; a is 5.

Answer and reasoning

The setup is right, but $\sin 30^\circ = 0.5$, so $a = 5$. A negative side length is a sign that the calculator was in radian mode and took the sine of 30 radians. Switch to degree mode. **The correct value is $a = 5$.**

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Example 17: Slip 5

$$\tan A = 8/15$$
$$A = \tan(8/15) \approx 0.01$$

Question

A student wrote these lines. What is the mistake, if any?

A. To get the angle back, use inverse tangent: about 28.1°.
B. The ratio should be flipped to 15 over 8, giving about 61.9°.
C. There is no mistake.
D. The answer should be given in radians instead of degrees.

Answer and reasoning

Tangent takes an angle and gives a ratio. Here the ratio is known and the angle is wanted, so use inverse tangent: $A = \tan^{-1}(8/15) \approx 28.1^\circ$. **The correct value is $A \approx 28.1^\circ$.**